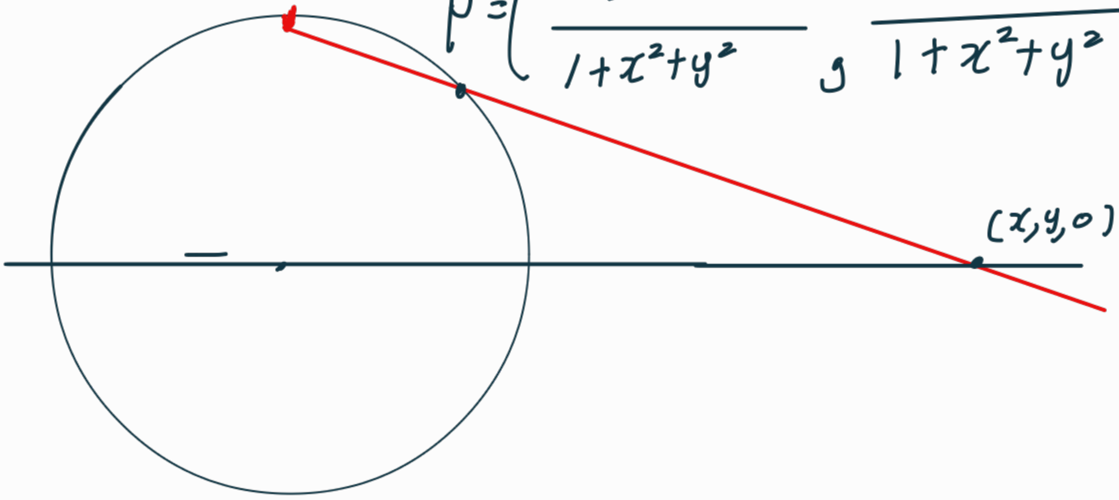
$$P = \left(\frac{2x}{1+x^2+y^2}, \frac{2y}{1+x^2+y^2}, \frac{x^2+y^2-1}{1+x^2+y^2} \right)$$



$$\langle P_x, P_x \rangle = \langle P_y, P_y \rangle = \frac{4}{1+x^2+y^2}$$

$$\langle P_x, P_y \rangle = 0.$$

먼저 이걸을 복소수를 이용하여 나타내자.

$z = x + iy$ 라 두면, $\bar{z} = x - iy$

$dz = dx + i\,dy \quad d\bar{z} = dx - i\,dy.$

$f(z) = f(x+iy)$, $x, y \in \mathbb{R}$

$f(x+iy) = u(x,y) + i\,v(x,y)$, $\boxed{u, v \in \mathbb{R}^2 \to \mathbb{R}}$

$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$ 그리고 $dx = \frac{dz + d\bar{z}}{2}$

$dv = \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy.$ $\quad i\,dy = \frac{dz - d\bar{z}}{2}$

$$df = du + i\,dv$$

$$= \left(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}\right)\left(\frac{dz - d\bar{z}}{2}\right) + \left(\frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}\right)\left(\frac{dz + d\bar{z}}{2}\right) = \frac{\partial f}{\partial x}\left(\frac{dz + d\bar{z}}{2}\right) - i\frac{\partial f}{\partial y}\left(\frac{dz - d\bar{z}}{2}\right)$$

$$= \frac{1}{2}\left(\frac{\partial f}{\partial x} - i\frac{\partial f}{\partial y}\right)dz + \frac{1}{2}\left(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\right)d\bar{z}.$$

$$\left(\frac{1}{2}\left(\frac{\partial f}{\partial x} - i\frac{\partial f}{\partial y}\right) := \frac{\partial f}{\partial z},\ \frac{1}{2}\left(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\right) := \frac{\partial f}{\partial \bar{z}}\right)$$

$$df = \frac{\partial f}{\partial z}dz + \frac{\partial f}{\partial \bar{z}}d\bar{z}.$$

Example

$$d(z^n) = \frac{\partial z^n}{\partial z}dz + \frac{\partial z^n}{\partial \bar{z}}d\bar{z}.$$

$$= \left[\frac{1}{2}\frac{\partial z^n}{\partial x} - \frac{1}{2}i\frac{\partial z^n}{\partial y}\right]dz + \frac{\partial z^n}{\partial \bar{z}}d\bar{z}$$

$$= \left[\frac{n}{2}z^{n-1}\frac{\partial z}{\partial x} - \frac{n}{2}iz^{n-1}\frac{\partial z}{\partial y}\right]dz + \frac{\partial z^n}{\partial \bar{z}}d\bar{z}$$

$$= nz^{n-1}\left(\frac{\partial z}{\partial z}\right)dz + nz^{n-1}\left(\frac{\partial z}{\partial \bar{z}}\right)d\bar{z}$$

한편, $\frac{\partial z}{\partial z} = 1$, $\frac{\partial z}{\partial \bar{z}} = 0$ 이므로, $d(z^n) = nz^{n-1}dz$.

$$|dz|^2 = dz\,d\bar{z} = (dx + i\,dy)(dx - i\,dy)$$

$$= dx^2 + dy^2 + i\,dy\,dx - i\,dx\,dy$$

$$= dx^2 + dy^2.$$

$$\frac{\partial}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right)$$

$$+ \quad \frac{\partial}{\partial \bar{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)$$

$$\frac{\partial}{\partial z} + \frac{\partial}{\partial \bar{z}} = \frac{\partial}{\partial x} \quad \text{---}(*)$$

$$i\frac{\partial}{\partial z} = \frac{i}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right)$$

$$-i\frac{\partial}{\partial \bar{z}} = \frac{-i}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)$$

$$i\frac{\partial}{\partial z} - i\frac{\partial}{\partial \bar{z}} = \frac{\partial}{\partial y} \quad \text{---}(*)$$

$$\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = \left(\frac{\partial^2}{\partial z^2} + 2\frac{\partial}{\partial z}\frac{\partial}{\partial \bar{z}} + \frac{\partial^2}{\partial \bar{z}^2}\right)$$

$$+ i^2\left(\frac{\partial^2}{\partial z^2} - 2\frac{\partial}{\partial z}\frac{\partial}{\partial \bar{z}} + \frac{\partial}{\partial \bar{z}^2}\right)$$

$$= 4\frac{\partial}{\partial z}\frac{\partial}{\partial \bar{z}}.$$

$$A\frac{\partial}{\partial x} + B\frac{\partial}{\partial y} \overset{(*)}{=} A\left(\frac{\partial}{\partial z} + \frac{\partial}{\partial \bar{z}}\right) + iB\left(\frac{\partial}{\partial z} - \frac{\partial}{\partial \bar{z}}\right)$$

$$= \alpha\frac{\partial}{\partial z} + \bar{\alpha}\frac{\partial}{\partial \bar{z}}$$

Example

$\lambda(z) := \dfrac{4}{1+|z|^2}$ 또는 $\lambda(x,y) = \dfrac{4}{1+x^2+y^2}$.

$$\frac{\partial \lambda}{\partial x} = \frac{-8x}{(1+x^2+y^2)^2}, \quad \frac{\partial \lambda}{\partial y} = \frac{-8y}{(1+x^2+y^2)^2}$$

$$2\frac{\partial\lambda}{\partial x}+\frac{\partial\lambda}{\partial y}=\frac{-16x-8y}{(1+x^2+y^2)^2}$$

$$(2+i)\frac{\partial\lambda}{\partial z}+(2-i)\frac{\partial\lambda}{\partial\bar z}$$

$$=(2+i)\frac{(-4\bar z)}{(1+|z|^2)^2}\ (\because z\bar z)\ +(2-i)\frac{(-4z)}{(1+|z|^2)^2}$$

$$=\frac{-4}{(1+|z|^2)^2}\left(\underbrace{2(\bar z+z)+i(\bar z-z)}_{2(2x)+i(-2iy)=4x+2y}\right)$$

이제 $ds^2=\dfrac{4}{(1+x^2+y^2)^2}(dx^2+dy^2)$ 을 복소화합시다.

$\dfrac{4|dz|^2}{(1+|z|^2)^2}$ 으로 먼저 바꿔줍니다.

$$\left\langle\frac{\partial}{\partial x}P,\frac{\partial}{\partial x}P\right\rangle=\cdots=\frac{4(dx^2+dy^2)\left(\frac{\partial}{\partial x},\frac{\partial}{\partial x}\right)}{(1+x^2+y^2)^2}$$

$$=\frac{4|dz|^2\left(\underbrace{\frac{\partial}{\partial z}+\frac{\partial}{\partial\bar z}}_{\frac{\partial}{\partial x}},\underbrace{\frac{\partial}{\partial z}+\frac{\partial}{\partial\bar z}}_{\frac{\partial}{\partial x}}\right)}{(1+|z|^2)^2}$$

$$=\frac{4\,dz\otimes d\bar z\left(\frac{\partial}{\partial z}+\frac{\partial}{\partial\bar z},\frac{\partial}{\partial z}+\frac{\partial}{\partial\bar z}\right)}{(1+|z|^2)^2}=\frac{4}{1+|z|^2}.$$

$$\left\langle\frac{\partial}{\partial y}P,\frac{\partial}{\partial y}P\right\rangle=\cdots=\frac{4|dz|^2\left(-i\left(\frac{\partial}{\partial z}-\frac{\partial}{\partial\bar z}\right),-i\left(\frac{\partial}{\partial z}-\frac{\partial}{\partial\bar z}\right)\right)}{(1+|z|^2)^2}$$

$$= \frac{4\, dz \otimes d\bar{z}\left(-\frac{i}{2}\left(\frac{\partial}{\partial z}-\frac{\partial}{\partial \bar{z}}\right), -\frac{i}{2}\left(\frac{\partial}{\partial z}-\frac{\partial}{\partial \bar{z}}\right)\right)}{(1+|z|^2)^2}$$

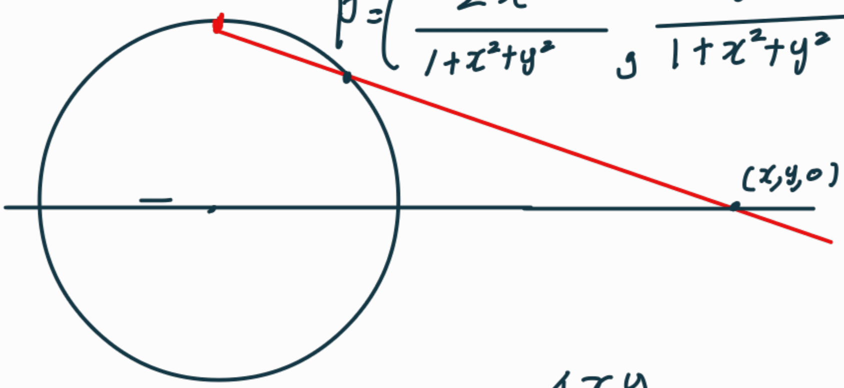
$$= \frac{4\left(-\frac{i}{2}\right)\cdot\left(\frac{i}{2}\right)}{(1+|z|^2)^2} = \frac{4}{(1+|z|^2)^2}.$$

좋다 그럼 애초에 $\frac{4}{(1+|z|^2)^2}$ 는 어떻게 얻지?

여태까지 $\langle \frac{\partial}{\partial x}P, \frac{\partial}{\partial x}P\rangle = \langle \frac{\partial}{\partial y}P, \frac{\partial}{\partial y}P\rangle$, $\langle \frac{\partial}{\partial x}P, \frac{\partial}{\partial y}P\rangle = 0$ 계산에서 쭉 출처 (공 $\subset \mathbb{R}^3$)

로 부터 내려오는 길이라면, ... 복소 $\mathbb{C}P^1$ 에서도

이걸을 얻을 수 있을까?



$$P = \left(\frac{2x}{1+x^2+y^2}, \frac{2y}{1+x^2+y^2}, \underline{\frac{x^2+y^2-1}{1+x^2+y^2}}\right)$$

$$= 1 - \frac{2}{D}$$

$$P_x = \left(\frac{2D-4x^2}{D^2}, \frac{-4xy}{D^2}, \frac{4x}{D^2}\right)$$

$$P_y = \left(\frac{-4xy}{D^2}, \frac{2D-4y^2}{D^2}, \frac{4y}{D^2}\right)$$

$$\langle P_x, P_x\rangle = \frac{4}{D^4}\left((D-2x^2)^2 + \underbrace{4x^2y^2 + 4x^2}_{4x^2(D-x^2)}\right)$$

$$= \frac{4}{D^4}\left(D^2 - 4x^2D + 4x^4 + 4x^2D - 4x^4\right) = \frac{4}{D^2}$$

$$\langle P_x, P_y \rangle = \frac{4}{D^4} \langle (D-2x^2, -2xy, 2x), (-2xy, D-2y^2, 2y) \rangle$$

$$= \frac{4}{D^4} \left[-2xyD + 4x^3y - 2xyD + 4xy^3 + 4xy \right]$$

$$= \frac{8xy}{D^4} \left[-D + 2x^2 - D + 2y^2 + 2 \right] = 0$$

$$\langle P_y, P_y \rangle = \frac{4}{D^4} \left(4x^2y^2 + (D-2y^2)^2 + 4y^2 \right) = \frac{4}{D^2}$$